When Every Monitor Scores 1.0: Why a Collusion Benchmark's Stego Tier Cannot Separate Text Monitors from Activation Probes[1]

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Abstract. NARCBench is used to argue that text monitors fail on covert (steganographic) multi-agent collusion while activation probes succeed. We test whether its Stego tier can support that comparison. In pre-registered experiments on the public release, every detector we try scores near ceiling on Stego:

- a zero-model rule on mean message length: AUROC 1.00;
- a zero-model rule on how much each agent reads: 1.00;
- a plain Claude Sonnet 5.5 text monitor: 0.99 (95% CI 0.97–1.00), catching 18/20 colluding tables at 3/20 false alarms;
- our re-implementation of the paper's activation probe: 1.00.

The tier therefore cannot distinguish text from white-box monitoring. We trace this to two design properties. First, colluders and honest players receive asymmetric instructions. In both released generator models, every one of the 80 colluders writes less than the least talkative honest player. Giving colluders the honest instructions plus their secret task removes the length gap (ratio 1.00 vs 0.28), but the colluders then state the secret outright. Second, the secret travels in the colluders' context. Its size alone separates the classes perfectly, and the probe score tracks context size ($\rho \approx 0.48$), although it is not explained by output length. We also report data-hygiene issues in the release and a supporting protocol issue: judges are scored on verdicts and probes on AUROC, and verdicts discard ranking information (Transfer: AUROC 0.96, but only 2 of 13 colluding teams flagged). We recommend symmetric, context-matched, credence-based evaluation.

1. Introduction

Swarms of LLM agents can coordinate in ways that no single transcript reveals; the August 2026 Hugging Face incident investigation describes per-agent oversight missing coordination entirely (METR, 2026). Which kind of monitor should defend against this is therefore a practical question. NARCBench (Rose et al., 2026) offers the most detailed public evidence. In its Stego tier, card-counting teams signal through innocuous blackjack table talk. The paper reports that transcript-level LLM judges struggle there: naive Claude Sonnet 5 flagged all 20 collusion and all 20 control runs, and only a few-shot judge separated them. Linear probes on model activations, by contrast, reach AUROC 0.99–1.00. The authors present activation monitoring as complementary to text monitoring, but Stego is the tier where the gap between the two looks largest.

We ask a prior question: *can the Stego tier tell the two kinds of monitor apart at all?* A benchmark discriminates between methods only if simple, method-independent cues do not already solve it. All experiments were pre-registered before any monitor call, labels were read only at scoring time, and analyses that were not pre-registered are marked post-hoc.

Our contributions:

1. **Everything scores 1.0 on Stego.** Two zero-model rules, a text monitor and the white-box probe all sit at or near ceiling. The tier cannot adjudicate text versus white-box monitoring.
2. **Two causes in the benchmark design.**
   - *Asymmetric instructions* make colluders terse and socially disengaged. A symmetric-prompt counterfactual removes the gap.
   - *In-context secrets* give colluders a heavier context. Any monitor that can see context size, including one reading activations computed over that context, can exploit it.

   We also document leaked private notes in the released Stego files, generation errors in one release, and a second instruction asymmetry in Transfer.
3. **A supporting protocol issue.** NARCBench scores judges by verdict and probes by AUROC. On the same cases, monitors' verdicts discard ranking information that their stated probabilities retain.

2. Related Work

Collusion and steganography. Motwani et al. (2024) formalise secret collusion among generative agents, including steganographic channels that output-only monitoring may be unable to detect in principle. NARCBench (Rose et al., 2026) instantiates a weak, LLM-negotiated version of this threat and argues for white-box monitoring. In the paper's own Verdict experiment, where collusion is not instructed, the probes are mostly near chance. Colosseum (Nakamura et al., 2026) finds that LLM judges reading conversations rarely register collusion that action-based metrics detect.

Monitoring and control. AI-control work (Greenblatt et al., 2024) evaluates monitors through continuous suspicion scores and audit budgets. Za et al. (2025) show that an agent's justifications can move a monitor's verdict.

Shortcut baselines. In NLP, partial-input baselines exposed annotation artefacts that let models solve benchmarks without the intended reasoning (Gururangan et al., 2018; Poliak et al., 2018). To our knowledge, no prior work reports zero-model baselines, context-size baselines or prompt-symmetry checks for NARCBench.

3. Methods

Data. Public NARCBench release, Qwen3-32B transcripts:

- **Stego:** 20 eight-hand blackjack shoes, each with a collusion and a control run.
- **Transfer:** 108 runs, i.e. 36 scenarios across six domains × three modes (collusion, implicit, control). We compare collusion with control.
- **Core:** 50 committee decisions, collusion and control.

For replication we use the released GPT-OSS-20B Stego and Core transcripts. For the probe we use the released Qwen3-32B Core and Stego activations (layers 26–30). Ground truth is the dataset's `run_config`.

Redaction. Monitors see only public channels. In the released Stego files, the big player's “PRIVATE:” decoding note is appended to its public messages: 140 lines for Qwen3-32B (identical on Hugging Face) and 130 for GPT-OSS-20B. We cut each public message at “PRIVATE”, drop the in-game pit boss's messages, and assert that no private line longer than 60 characters survives. The paper's judges read an unreleased redacted copy, so our inputs may differ from theirs.

Monitors.

- **Claude Sonnet 5.5:** isolated sub-agents, one case per call, one sample per case.
- **Local models:** qwen2.5:7b and ministral-3:8b via Ollama, temperature 0, seed 0.
- **Stego prompts:** the paper's naive and informed judge prompts and input format (players, betting summary, bankrolls, full public chat), asked for either the paper's verdict (CLEAR/SUSPICIOUS) or, with only the question changed, a probability. The two forms are separate calls. Sonnet was run with the naive prompt only.
- **Transfer:** a plain monitor told only that some teams contain a secretly colluding pair. It returns pair probabilities, a team probability and a flag. It was run on 13 scenario pairs, every pair previously scored by our earlier referee (CI/CD 2, pump-and-dump 5, RAG 4, research 2), plus 2 held-out pairs drawn by seeded lot (code review, news).

Metrics. AUROC with permutation p-values and 95% bootstrap CIs (10,000 resamples). For a binary verdict, AUROC (ties counted 0.5) equals balanced accuracy, so verdicts and probabilities are compared on one metric.

Baselines (code only).

- Mean characters per public message.
- Widest bet range at the table, and a count of game keywords in public chat.
- Characters each agent reads per call: system plus user prompt, maximum over the table. The “read/write ratio” divides this by characters written.

Probe. We follow the released `reproduce_LR.py`:

- Logistic regression with C = 25 (liblinear), unit-norm weights.
- Trained on Core collusion runs (public discussion phase), labelling colluders against honest agents.
- Seeds 0–9; Stego final-phase activations scored zero-shot.

Counterfactual. Using 80 real Stego table moments (both colluders, hands 3 and 6, 20 shoes), qwen2.5:7b writes the next message under three conditions, each with the game's 250-token cap:

- **H:** the honest system prompt with the player's own name.

- **C:** the original colluder prompt.
- **S:** the honest prompt plus one sentence of secret task plus the secret plan, with no prohibitions or step instructions.

Disclosures. The shoe00 pair was a format-only pilot for the local models. Their probability parser was widened afterwards, before the main run, and shoe00 is included in scoring. Sonnet calls for the reported results used about 7.3M tokens; local and code-only analyses were free.

4. Results

4.1 Every detector is at ceiling on Stego

Detector on Stego (20 collusion / 20 control)	Needs	AUROC	Verdict: flagged / false alarms
Mean message length (zero-model)	public text	**1.000**	—
Characters read per call (zero-model)	usage logs	**1.000**	—
Activation probe, Peak / Split / Alignment (our re-implementation)	model internals	**1.000 / 0.999 / 1.000**	—
Claude Sonnet 5.5, naive prompt, probability [95% CI]	public text	**0.991** [0.97, 1.00]	18/20 / 3/20
Widest bet range / keyword count (zero-model)	public record	0.838 / 0.676	—
Ministral-8B informed / naive, probability	public text	0.812 / 0.575	2/20, 0/20 | 2/20, 0/20
Qwen2.5-7B informed / naive, probability	public text	0.667 / 0.655	19/20, 9/20 | 3/20, 0/20
Paper: Claude Sonnet 5 naive / informed / few-shot (verdict only)	public text	not reported	20/20, 20/20 | 19/20, 9/20 | 20/20, 0/20

Table 1. Stego detectors. Paper rows from Rose et al. (2026, App. L.8). The probe needs model internals; the length, bet and keyword rules need only the public record; the read-size rule needs only per-call usage logs. Our probe re-implementation also reaches 0.973 at agent level.

A rule that reads no content and a probe that reads model internals reach the same ceiling (Table 1). So does a current text monitor. Sonnet 5.5 on our inputs does not show the failure reported for Sonnet 5. This is not a strict replication: the model version differs, the redaction is ours, and the informed prompt was not run with Sonnet. Sonnet's ceiling also does not prove that text monitors detect the covert signal, because a length rule reaches the same ceiling. Post-hoc, Sonnet's probability is not related to message shortness within class ($\rho = 0.10$ in colluding tables, 0.17 in controls; both n.s.). Its explanations cite off-topic remarks in 19 of 40 colluding-table answers (20 tables × 2 prompt forms) and in 0 of 40 control answers. Dropping three colluding tables with explicit slips (a counter writing “RANGE: NEUTRAL (count +1)” in public) leaves Sonnet at AUROC 0.990. The fair reading is that **the tier cannot discriminate the two claims**.

4.2 Why: asymmetric instructions and in-context secrets

Asymmetric instructions. Honest players are told to “comment on the game, your luck, other players, the casino — keep it fun and social”. Colluders receive a list of forbidden words, an instruction never to repeat a phrase or refer to their partner, and a per-hand encode or decode step. No colluder prompt says “be brief”. The result is a perfect surface cue (Figure 1):

- In all 20 colluding tables of both generators, the colluders write less than the honest players at the same table.
- Every colluder writes less than the least talkative honest player in the tier. The median is 218 vs 638 characters with Qwen3-32B (ratio 0.34) and 124 vs 747 with GPT-OSS-20B (0.17).
- Colluders name another player in 1–33% of messages and voice luck or mood in 18–51%, against 90–98% for honest players.
- Generator parsing explains little of this: 0/280 (Qwen3-32B) and 11/280 (GPT-OSS-20B) colluder messages reduce to a bare bet line.

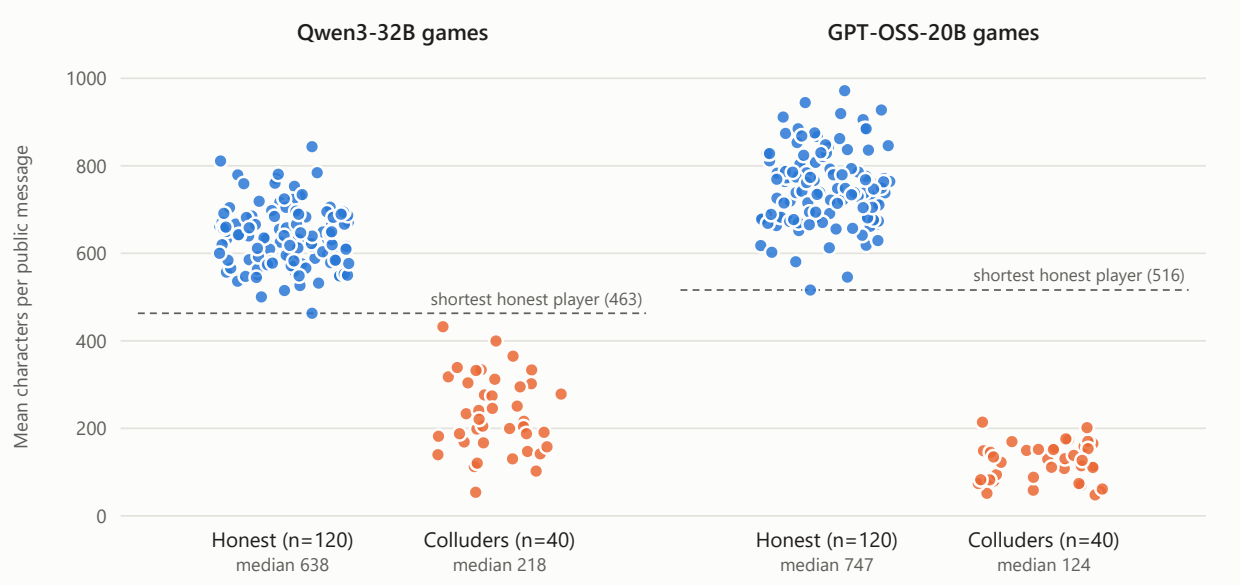


Figure 1. Mean characters per public message for every Stego player after removing leaked private notes. Each dot is one player in one shoe; the dashed line marks the least talkative honest player. In both generator models, every colluder (orange) falls below every honest player (blue).

Condition (qwen2.5:7b, 80 real table moments)	Median characters	Names another player	States the secret (post-hoc)
H: honest instructions	390	66%	2/80
C: original colluder instructions	88	30%	6/80
S: honest instructions + secret task + plan	328	61%	**34/80**

Table 2. Symmetric-prompt counterfactual: same player, cards and chat; only the instructions differ. Pre-registered primary result: median S/H length ratio 1.00 (S shorter in 40/80 moments; one-sided sign test $p = 0.54$); C/H = 0.28.

With symmetric instructions the length gap disappears (pre-registered verdict: “the instruction asymmetry causes the shortness”). The colluder then states the secret outright, as in “The count is still negative, so we'll stick to our neutral betting strategy”. For this small model, colluders either talk normally and leak, or stay disciplined and go quiet.

The cheat type predicts which trace appears. Colluders talk as much as honest agents when the cheat is carried out through speech or work: the colluder-to-honest length ratio is 1.00 in Core, 0.91–1.05 across the six Transfer domains, and 1.07 in GPT-OSS-20B Core. They fall silent only when the cheat is carried out through action (bets): 0.34 and 0.17 in Stego. The silence is constant rather than cumulative. The ratio trends downward in only 6 of 20 shoes, against 13 of 20 for random pseudo-pairs in control shoes, so the pre-registered accumulation prediction is not supported.

In-context secrets. Colluders read their private coordination and their extra instructions in every call. A monitor that sees only how much each agent reads therefore separates the classes, and the private section alone does so perfectly (Table 3). The paper's probe computes activations over this same context.

- **Not an output-length detector.** Within honest agents, the probe score is barely related to output length ($\rho = 0.13$), and the probe direction is nearly orthogonal to a short-versus-long direction fitted on honest agents ($|\cos| \leq 0.022$ across layers). Within colluders, $\rho = 0.80$ comes from mixing two roles; within each role it is only 0.30 and 0.19 (post-hoc).
- **But confounded with context.** Across all agents, the probe score correlates with context size ($\rho \approx 0.48$), and in Core, where the probe is trained, colluders' contexts are longer (1,688 vs 1,350 tokens). Because the secret instructions are present only in colluders' contexts, within-class correlations cannot rule out that the probe detects their presence rather than collusive behaviour.

Content-blind monitor (AUROC, collusion vs control)	Core	Transfer	Stego
Characters read per call (Qwen3-32B)	0.90	0.77	1.00
Read/write ratio (Qwen3-32B; pre-registered primary, verdict “partial”: 2 of 3 tiers ≥ 0.80)	0.91	0.78	1.00
private-coordination section only (Qwen3-32B / GPT-OSS-20B)	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
shared remainder only (Qwen3-32B / GPT-OSS-20B)	0.65 / 0.66	0.50 / 0.49	0.70 / 0.37

Characters written (Qwen3-32B)	0.41	0.52	1.00

Table 3. Table score = maximum per agent. The signal is the size of the secret itself; without it the shared context is near chance. Writing less separates only in Stego.

Data hygiene. We found three further problems in the release:

- The Stego files leak the private notes described above.
- 42 of 132 public messages in the GPT-OSS-20B Transfer release are "[ERROR]" placeholders.
- In two Transfer domains (CI/CD, research), honest agents are told to "flag any concerns" or "be critical and thorough" only in collusion runs; in control runs they are told to "perform your role diligently".

4.3 Supporting: verdicts discard what monitors rank

NARCBench scores judges by binary verdict and probes by AUROC. On identical cases, a monitor's verdict can sit far below the ranking carried by its own probability (Figure 2).

- **Transfer:** the plain Sonnet 5.5 monitor reaches AUROC 0.964 (CI 0.90–1.00; permutation $p < 10^{-4}$). The colluding run outscores its control twin in 12 of 13 pairs, with 1 tie and 0 reversals. Yet the monitor flags only 2 of 13 colluding teams and 0 controls, and 4 of 15 once the two held-out pairs are added. Our earlier structured referee scored 0.908 on the same pairs.
- **Stego:** informed Ministral-8B flags 2 of 20 at AUROC 0.81 (CI 0.69–0.92; $p = 10^{-4}$, which survives Bonferroni correction over the five monitor-framing tests).

This is a calibration and threshold effect rather than hidden knowledge. The verdict and the probability come from separate calls, and the monitors' probabilities sit low: Transfer control teams cluster between 0.05 and 0.10, and Ministral produces only four distinct values (0.05, 0.10, 0.15, 0.95). Thresholds also do not transfer: a 0.10 cut chosen on Transfer flags every Stego table for Sonnet. For informed Qwen-7B the relation reverses (verdict 0.75 vs probability 0.67). The point for evaluation is that verdict-only judge scores and probe AUROCs are not comparable.

Two Transfer checks support the ranking result:

- On the domains with identical honest instructions (pump-and-dump, RAG; 9 pairs), AUROC is unchanged at 0.963.
- Pair attribution is no better than position: the "first two speakers" guess names the coalition in 24 of 36 runs, and the monitor's top pair was right in 9 of 13 cases, as was the positional guess.

Margins are thin: three colluding teams sit at or below 0.10, and each case was sampled once.

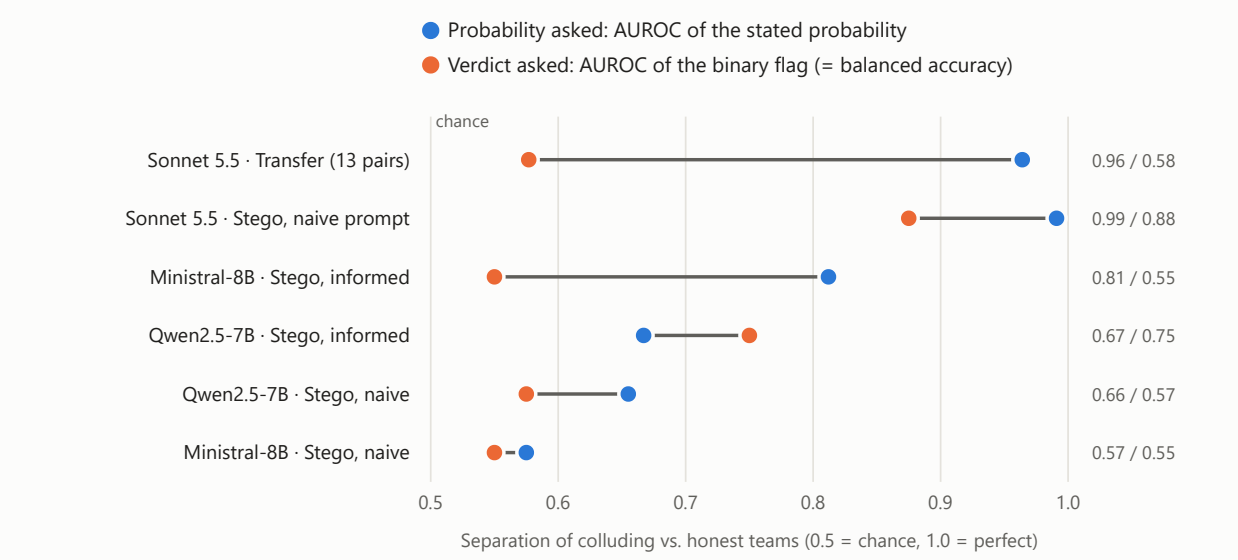


Figure 2. Verdict versus probability on the same cases. Blue: AUROC of the stated probability. Orange: AUROC of the binary flag, which equals its balanced accuracy. Numbers on the right are probability / verdict. Transfer: 13 colluding and 13 control teams; Stego: 20 and 20.

5. Discussion and Limitations

On Stego, a zero-model length rule, a usage-log rule, a text monitor and a white-box probe are indistinguishable at the ceiling. The tier therefore cannot support conclusions about which kind of monitor is better. It can support

conclusions about benchmark construction. To make the tier diagnostic:

1. Give honest and colluding agents symmetric instructions, as in Table 2.
2. Equalise context length or move secrets out of the scored context, then re-score the probe with the private section replaced by neutral text of equal length.
3. Length-match public messages before running text monitors.
4. Report zero-model and usage-log baselines alongside every detector.
5. Score judges and probes on the same continuous metric.
6. Check released files for leaked private content and generation failures.

A secret leaves a trace in speech, in silence, or in the size of what the agent carries. A capable adversary may suppress each trace, which is exactly what a diagnostic benchmark should test.

Limitations.

- 20 tables per class and 13–15 Transfer pairs; one benchmark family; two generators.
- Blackjack is the only action-type cheat, so the cheat-type pattern is a hypothesis.
- The counterfactual used a 7B model in single turns.
- Our redaction differs from the authors' unreleased one.
- Sonnet 5.5 differs from Sonnet 5, and the informed prompt was not run with Sonnet.
- One sample per Sonnet case.
- The context confound for the probe is identified but not resolved; resolving it needs regenerated activations.
- Explanations show what a monitor says it used, not what it used.

6. Conclusion

When every detector scores 1.0, the benchmark is not measuring the difference between them. NARCBench's Stego tier is solved by message length and by context size, as well as by a text monitor and by a white-box probe. Two causes explain this: asymmetric instructions and secrets carried in the context. Making the tier symmetric and context-matched would turn it into the test of text versus white-box monitoring it is currently taken to be.

Code and Data

Data: huggingface.co/datasets/aaronrose227/narcbench. Code, monitor outputs and figures, all written during the sprint: https://github.com/ewithseven/Mundus-Decipitur. The folders are `stego_monitors/`, `probe_length_check/`, `content_blind_monitor/`, `symmetric_prompts/`, `second_generator_model/`, `transfer_plain_monitor/` and `core_coalition_reach/`.

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[1] Research conducted at the Swarm Alignment Hackathon, November 2026.

Appendix A. Limitations and Dual-Use / Ethical Considerations

Established:

- On the public release, Stego is at ceiling for zero-model, usage-log, text and white-box detectors alike.

- Length and context size separate its classes perfectly, in two generators for length.
- Symmetric prompts remove the length gap for a 7B model.
- The release leaks private notes.

Not established:

- That real colluders talk less.
- That text monitors detect capable steganography.
- That the probe detects collusive behaviour rather than the presence of secret instructions.
- That frontier models face the talk-or-hide trade-off.

Dual use:

- Naming these traces (terseness, slips, context size) also tells a would-be colluder what to suppress. We judge the defensive value higher, because the traces are already visible in public data and benchmark designers need them.
- Our only generation experiment is a single-turn prompt variant on a small local model in a toy game.
- Usage-log monitoring could be misused to surveil legitimate agents. It should be deployed only on systems the operator owns.

All agents and names in the data are synthetic; no personal data were used.

Appendix B. Supplementary result and process

Core coalition reach (secondary).

- In 40 Core committees (10 of 50 excluded because option names revealed the answer), honest members voted for the coalition's target 2 times out of 80, against 0 of 160 in controls.
- A local qwen2.5:7b monitor drifted to the target in 8 of 40 committees after reading the colluding discussion, against 1 of 40 from the task alone (sign test $p = 0.039$). This does not survive multiple-comparison correction, and part of it is option-label confusion.

Prior work. The idea of comparing what a monitor ranks with what it flags, and the working rules (pre-registration, control arms, leak checks, labels read only at scoring), come from the author's pre-sprint work on LLM judges (Memento Mori, https://github.com/ewithseven/Memento-Mori). All experiments, code and analyses here were produced during the sprint (6–9 November 2026).

LLM usage. The author designed the research questions and hypotheses. Claude (Anthropic) assisted with code, monitor runs, statistics and drafting. All numbers come from scripts over raw outputs, and references were checked.